

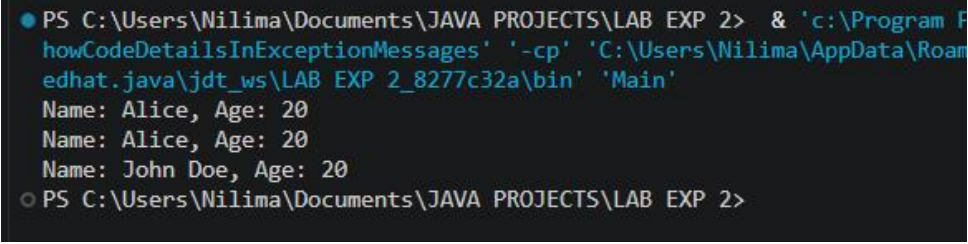
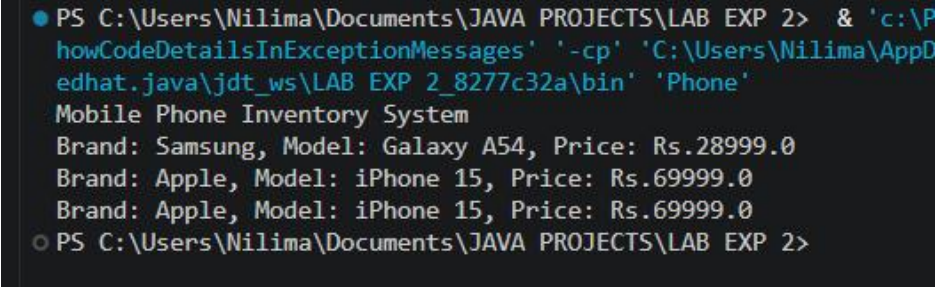


Symbiosis Institute of Technology, Pune

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No.: 2	
Name of Student	Akshit Gawade
PRN No.	25070122029
Batch	2025-2029
Class	A1
Academic Year & Semester	Second Year Semester III
Date of Submission	
Title of Assignment:	Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.
Problem Statement Learning Objectives	Problem Statement Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management. Learning Objectives To understand and implement default, parameterized, and copy constructors for object initialization.
Theory	Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Output Screenshots	 <pre> PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 2> & 'c:\Program Files\Java\jdk-8.0.602\bin\java.exe' -cp 'C:\Users\Nilima\AppData\Roaming\Microsoft\Windows\CurrentVersion\Shell\java\jdt_ws\LAB EXP 2_8277c32a\bin' 'Main' Name: Alice, Age: 20 Name: Alice, Age: 20 Name: John Doe, Age: 20 PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 2> </pre>
Exercise Qns	1. Create a Student class using default and parameterized constructors to initialize student name and roll number. 2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.
Output Screenshots	Qn 1 completed in main output. Qn 2:  <pre> PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 2> & 'c:\Program Files\Java\jdk-8.0.602\bin\java.exe' -cp 'C:\Users\Nilima\AppData\Roaming\Microsoft\Windows\CurrentVersion\Shell\java\jdt_ws\LAB EXP 2_8277c32a\bin' 'Phone' Mobile Phone Inventory System Brand: Samsung, Model: Galaxy A54, Price: Rs.28999.0 Brand: Apple, Model: iPhone 15, Price: Rs.69999.0 Brand: Apple, Model: iPhone 15, Price: Rs.69999.0 PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 2> </pre>
Conclusion	This assignment demonstrated how default, parameterized, and copy constructors enable flexible object initialization and secure data management in Java. The exercises proved that custom copy constructors prevent memory reference conflicts, ensuring independent data replication. Ultimately, mastering these constructor methodologies reinforces core OOP principles, directly improving code reliability and maintainability.